

Functional Requirements

Authentication & Authorization

- FR1: The system must allow registration of a new teacher or student account with a unique email per table.
- FR2: The system must hash passwords using PBKDF2 before storage.
- FR3: The system must provide login with role selection (Teacher/Student) using Cookie Authentication.
- FR4: The system must prevent any user from accessing pages that don't belong to their role (role-based authorization).

Course & Content Management (Teacher)

- FR5: A teacher can add/edit/delete a course, provided they own that course.
- FR6: A teacher can add classes belonging only to their own courses, and videos belonging only to their own classes.
- FR7: A teacher can create a quiz per class and an exam per course and link them to questions.
- FR8: Every question must have at least 2 choices with exactly one marked as correct.

Catalog & Enrollment (Student)

- FR9: The system must show students only courses open for enrollment (Enrollable = true).
- FR10: A student can enroll in a course only once (duplicate enrollment prevented).
- FR11: A student can access content only for courses they are enrolled in.

Assessments

- FR12: The system must auto-grade a student's answers on submission and compute the score out of the total number of questions.
- FR13: The system must store each attempt (score, timestamp) linked to the student and the assessment.
- FR14: The system must show per-question feedback (correct/incorrect) after submission.

Profile Management

- FR15: Any user (teacher or student) can edit their personal information.
- FR16: Any user can change their password after correctly entering their current password.